

Fitting the luminosity-function shape

Conditional likelihoods with and without measurement error in flux and distance

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Abstract

This note records how we fit the *shape* of a TNO luminosity function (LF) from a debiased pencil-beam survey, and why the likelihood looks the way it does. The fit described here is deliberately *shape only*: it does not use the Horvitz–Thompson sum $\sum 1/P$ as a Poisson mean. Absolute normalisation is handled separately. I first write down the no-measurement-error case (catalogue H treated as truth), then the version that folds in photometric scatter from the survey’s `.eff` file and, when available, per-object distance uncertainty. The aim is that someone who is comfortable with LF work, but not with quadrature jargon, can see what is being averaged over and what is not. Figures are generated by `docs/lf_shape_fit/make_figures.py`. A short results section applies the same machinery to JWST Sample A.

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1 What problem we are solving

A survey hands us a list of detections with apparent magnitudes and (approximate) distances. From those we form absolute magnitudes H , and we want the differential luminosity function

$$\lambda(H) \equiv \frac{dN}{dH} \quad (1)$$

of the underlying population. Two things get in the way:

1. **Selection.** Faint objects are less likely to be found. The survey efficiency, tracking fraction, and characterisation cut are known functions of apparent magnitude (and sometimes rate of motion). Call the product of those the selection function $S(m)$.
2. **Measurement error.** The magnitude we wrote in the catalogue is not the true magnitude, and the distance we used to turn m into H is not exact. Both move the inferred H .

This note is about fitting the *shape* of $\lambda(H)$ —parameters such as the bright-end slope α and a faint-end taper—while conditioning on the fact that each object was detected. We are not, in this document, fitting the total number of objects in the belt. That normalisation is a separate Horvitz–Thompson (or later, extended Poisson) step.

The functional form we use is the usual OSSOS-style tapered exponential:

$$\ln N(< H) = (\ln 10) \alpha (H - H_o) - 10^{-\beta(H-H_B)}, \quad (2)$$

with $\lambda(H) = dN/dH$. Here α is the logarithmic slope, H_B is where the taper sets in, and β controls how sharp the taper is. H_o is a reference magnitude that cancels in a shape-only fit; we keep it only so that the cumulative form matches the literature.

2 Why a conditional likelihood

A natural first instinct in LF work is a weighted histogram or a weighted likelihood $\sum_k w_k \log f(H_k | \theta)$ with $w_k = 1/P_k$. That is fine for some purposes, but it mixes shape and normalisation, and it treats $\sum w_k$ as if it were an independent Poisson count. For the shape of $\lambda(H)$ I prefer a likelihood that asks a narrower question:

Given that object k was detected, and given its geometry (distance, field, rate), what is the probability density of the magnitude we observed?

That is a *conditional* density. Write D_k for the event “detected in this survey under geometry k ”. The contribution of detection k is

$$f_k = p(\text{observed magnitude of } k \mid D_k, \text{geometry}_k, \theta). \quad (3)$$

The joint shape likelihood is $\prod_k f_k$. Absolute rate information has been conditioned out: two surveys with the same efficiency curve but different areas give the same shape likelihood for the same detections.

This is the same philosophy as a detection-conditioned LF fit in a magnitude-limited sample: the selection function appears in both the numerator and the denominator, and the overall survey yield cancels.

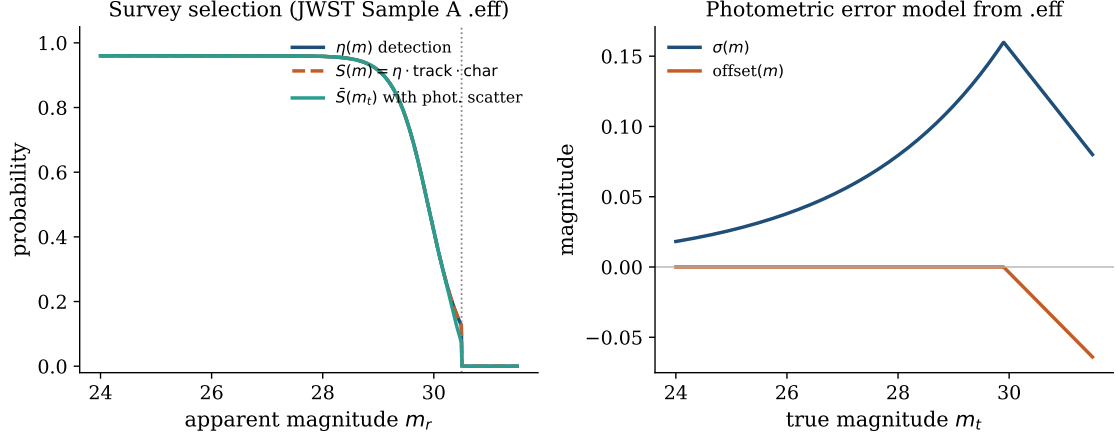


Figure 1: Left: JWST Sample A selection in the r -mapped band. $S(m)$ is the no-scatter product $\eta \cdot \text{track} \cdot \text{char}$. $\tilde{S}(m_t)$ (Section 4) averages tracking and characterisation over the photometric noise. Right: the `mag_error` law from the same `.eff` file—scatter $\sigma(m)$ and the faint-end magnitude offset.

3 No measurement error

3.1 Setup

Ignore photometric and distance error for a moment. Then each detection has a known true apparent magnitude m_k and a known true absolute magnitude

$$H_k = m_k - \mu_k, \quad (4)$$

where μ_k collects the distance modulus, colour term, and phase correction. (For the JWST Sample A numbers in this repo, μ_k is built from $m_r = m_{F150W2} + 1$ and $r = \Delta = d_{\text{bary}}$ with the Bowell $G = -0.12$ law, matching `grid_bias.apparent_to_Hr`.)

Detection probability at magnitude m is the selection function

$$S(m) = \eta(m) \text{track}(m) \text{char}(m), \quad (5)$$

read from the block’s `.eff` file: η is the dig probability, `track` is the tracking fraction, and `char` is the characterisation cut (`mag_lim`, or $\eta \geq 0.4$ when no limit is set).

3.2 Likelihood

Under a population $\lambda(H)$, the joint density of “draw an object of absolute magnitude H and detect it at this geometry” is proportional to $\lambda(H) S(H + \mu_k)$. Conditioning on detection replaces that with the normalised density

$$f_k(H_k) = \frac{\lambda(H_k) S(m_k)}{\int \lambda(H) S(m_k + H - H_k) dH}. \quad (6)$$

The factor $S(m_k)$ in the numerator does not depend on the LF parameters, so it drops out of a maximum-likelihood fit; what matters is that the *denominator* reweights $\lambda(H)$ by the selection function shifted onto the H axis for that object’s geometry.

What this buys us. Equation (6) is exactly what you want if the catalogue magnitudes really are truth. It is also the regression check for the measurement-error generalisation: when the photometric model is turned off (`--measurement none` in the fit CLI), the code must recover (6) to machine precision.

What this ignores. Near the efficiency drop, a 0.2–0.3 mag photometric error is not a small perturbation. Treating the catalogue H as exact then pulls the faint end of the LF in the wrong direction, because scattered-up faint objects are more common than scattered-down bright ones when $\lambda(H)$ is rising. That is the usual Eddington-style bias; Section 4 puts it in the likelihood instead of hoping the binning washes it out.

4 Photometric measurement error

4.1 What the survey simulator actually does

OSSSSim does not draw a Gaussian magnitude error and stop. For each plant it draws a measured magnitude from the `magran` model (`numutils.f95` / `ossssim.magerr`):

$$\hat{m} = m_t + \text{offset}(m_t) + \sigma(m_t) T_n, \quad (7)$$

where m_t is the true apparent magnitude, $\sigma(m)$ is a piecewise function of six `mag_error=` parameters, `offset(m)` is a deterministic faint-end shift (objects fainter than a threshold are measured systematically brighter when the last parameter is negative), and T_n is the mean of $n \in \{1, 2, 3\}$ unit-variance triangular random variables. The integers n are drawn with probabilities `phot_frac` (one, two, or three independent measures averaged).

Two ordering facts matter for the likelihood, because they are how the simulator decides detection:

- **Detection efficiency η is applied to the true magnitude m_t .**
- **Tracking and characterisation are applied to the measured magnitude $\hat{m}$.**

If we want the fit to speak the same language as the bias simulation, the likelihood has to respect that split.

4.2 Latent true magnitude

With scatter, the catalogue value $\hat{m}_k$ is no longer a deterministic map from H . True magnitude $m_t = H + \mu_k$ is latent. The contribution of detection k becomes an integral over that latent magnitude (or equivalently over H):

$$f_k = \frac{\int \lambda(H) \eta(m_t) p(\hat{m}_k | m_t) dH \times \text{track}(\hat{m}_k) \text{char}(\hat{m}_k)}{\int \lambda(H) \bar{S}(m_t) dH}, \quad m_t = H + \mu_k. \quad (8)$$

Here $p(\hat{m}_k | m_t)$ is the `magran` density (the triangular mixture above), and the denominator uses the *expected* selection at true magnitude m_t ,

$$\bar{S}(m_t) = \eta(m_t) \mathbb{E}[\text{track}(\hat{m}) \text{char}(\hat{m}) | m_t]. \quad (9)$$

The expectation is over the same photometric noise that the simulator draws. For a multi-epoch block each epoch draws its own noise, so $\bar{S}$ is a product over epochs.

The factors $\text{track}(\hat{m}_k) \text{char}(\hat{m}_k)$ sit outside the H integral in the numerator: they are constants for a fixed catalogue magnitude, so they cancel between detections of equal characterisation and do not move the MLE. Numerically we tabulate

$$\text{NUM}(H) = \eta(m_t) p(\hat{m}_k | m_t), \quad \text{DEN}(H) = \bar{S}(m_t) \quad (10)$$

on a fine H grid and replace the integrals by trapezoidal matrix products against $\lambda(H)$.

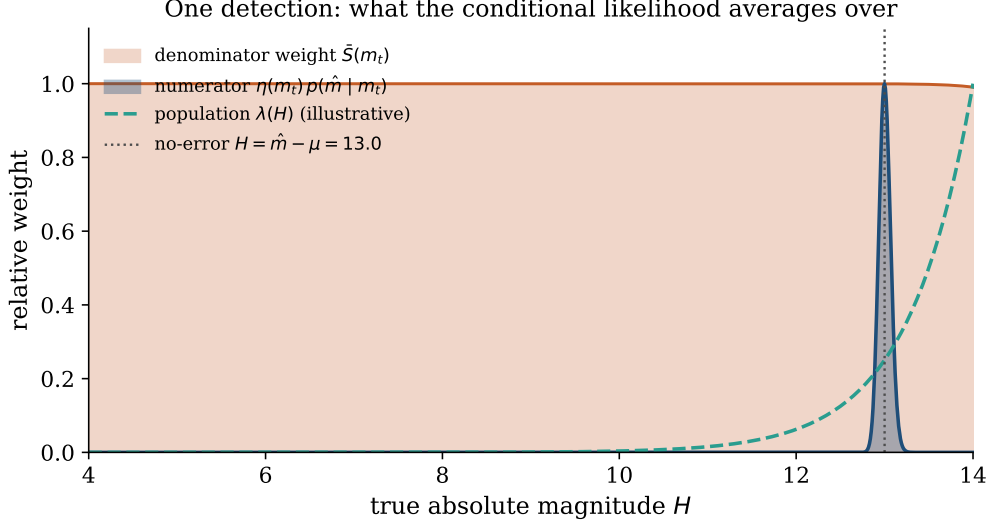


Figure 2: Cartoon for one detection. The blue curve is the numerator kernel: which true H values are consistent with the measured magnitude, after folding through η and the photometric model. The orange band is the denominator weight $\tilde{S}$. The dotted line is the no-error assignment $H = \hat{m} - \mu$. The green dashed curve is an illustrative population $\lambda(H)$. The conditional likelihood is the λ -weighted average of blue, divided by the λ -weighted average of orange.

4.3 Why this form, and not a simpler one

A few alternatives and why we did not stop at them:

Gaussian convolution of the LF. Blurring $\lambda(H)$ with a fixed σ_H is the right move when the noise is on H and selection is flat. Here the noise is on m , selection is a sharp function of m , and η and char see different magnitudes. A single convolution does not capture that.

Forward-model each detection with a Monte Carlo. Drawing many `magran` realisations per likelihood call works, and we use that as a unit test, but it is too noisy and too slow inside MCMC. The grid + analytic mixture kernel is the same integral with stable matrix algebra.

Put a prior on H_k and sample every object. Fully hierarchical Bayes is attractive and is not ruled out. For the shape-only product likelihood we need now, the per-object latent H integrates out in closed form on the grid, which keeps the fit comparable to the no-error code path and cheap to bootstrap.

The guiding constraint is consistency with OSSSSim: whatever we claim about bias in the HT weights, the shape fit should assume the same measurement process.

5 Distance uncertainty

Absolute magnitude depends on distance. If d is wrong, H is wrong. For short-arc detections the fractional distance error can be several percent, which is several tenths of a magnitude in the distance modulus when $r \sim \Delta \sim d$.

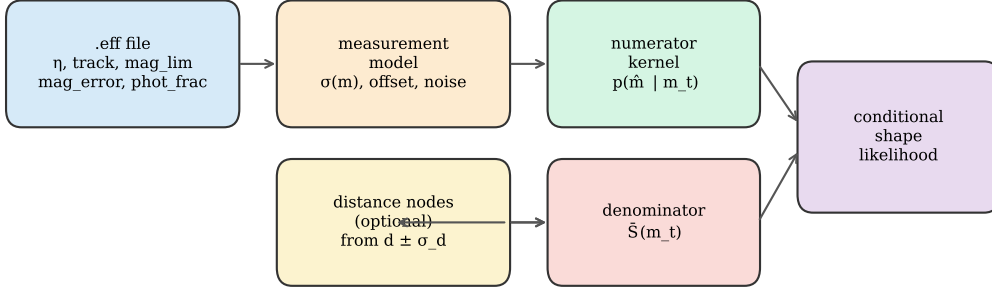


Figure 3: Information flow in the measurement-aware shape fit. The block `.eff` file supplies both the selection function and the photometric model. Distance nodes (Section 5) are optional and enter as a discrete mixture over geometries.

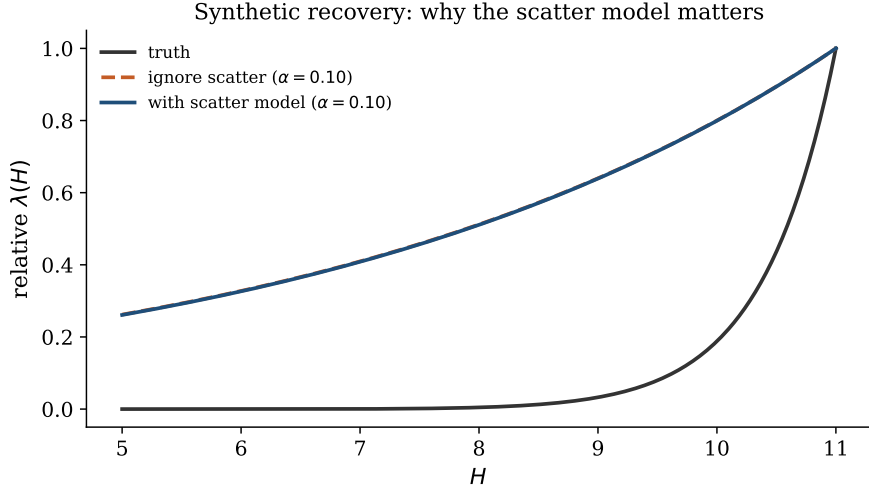


Figure 4: Synthetic check. Objects are drawn from a known tapered LF, detected with $\eta(m_t)$, and measured with the Sample A photometric model. Fitting while ignoring scatter (orange) returns the wrong shape; using the measurement model (blue) recovers the input (black).

5.1 What we assume

When a per-object distance error σ_d is available (column `d_bary_err` in the detection table), we treat the orbit-fit distance as a measurement of $\ln d$ with

$$\ln d \sim \mathcal{N}(\ln d_{\text{bary}}, [\ln(1 + \sigma_d/d_{\text{bary}})]^2). \quad (11)$$

That is: lognormal errors, fractional width σ_d/d . We are *not* yet folding in a population prior $p(d | i)$ from the OSSOS model; that is listed as future work. The present treatment is “the orbit fit said $d \pm \sigma_d$, and we had a flat prior in $\ln d$ before that measurement.”

5.2 How the integral is done (without the jargon)

The likelihood (8) now has one more latent variable: distance. We need to average f_k over the distance PDF. Rather than drawing thousands of random distances every time we evaluate the

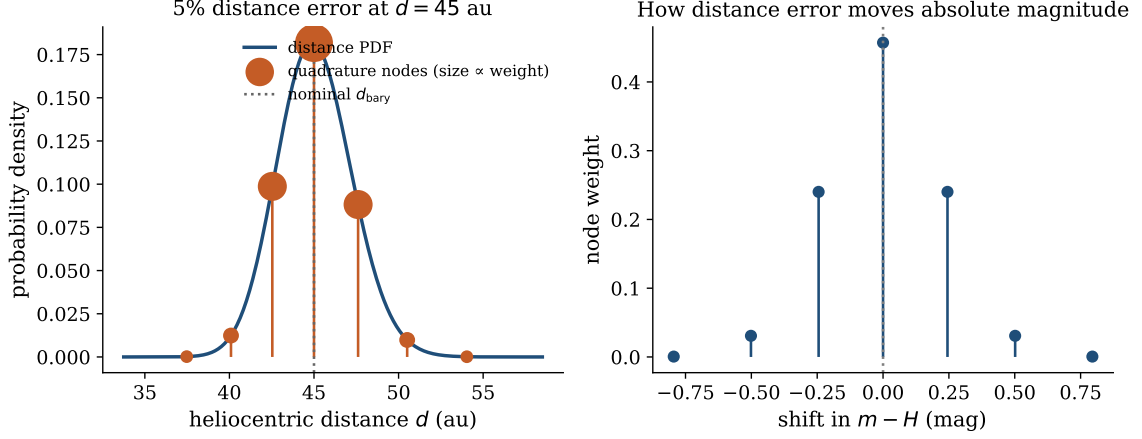


Figure 5: Left: lognormal distance PDF for $d = 45$ au with 5% fractional error, and the seven quadrature nodes (marker size $\propto$ weight). Right: the same nodes expressed as a shift in $m - H$. A 5% distance error is a ~ 0.2 mag uncertainty in absolute magnitude when $r \sim \Delta \sim d$.

likelihood, we replace the integral by a short weighted sum

$$f_k = \sum_{j=1}^J w_{kj} \frac{\text{NUM}_{kj}}{\text{DEN}_{kj}}, \quad (12)$$

where each index j is a single *representative distance* d_j , the weight w_{kj} is how much of the distance PDF that representative carries, and $\text{NUM}_{kj}/\text{DEN}_{kj}$ is equation (8) evaluated at that fixed distance (so μ_k becomes $\mu_k(d_j)$).

The particular distances and weights we use are the classical Gauss–Hermite quadrature nodes for a Gaussian in $\ln d$. In plain language: for a smooth integrand and a roughly Gaussian uncertainty, seven carefully chosen distances already reproduce a 200-point brute-force integral to a part in 10^3 or better. If there is no distance-error column, or if `--distance-nodes 1`, then $J = 1$, $d_1 = d_{\text{bary}}$, $w = 1$, and we are back to a fixed geometry.

5.3 What this is not

- It is not a joint orbit–LF fit. We do not re-fit a and e .
- It is not a hierarchical population distance model. Objects do not share a $p(d)$ that knows about the classical belt vs. scattering.
- It does not replace the HT cell assignment. Bias cells are still labelled by the catalogue H ; fixing that inconsistency is future work on the normalisation side.

6 Practical fitting

Grid. Everything is tabulated once on an H grid with step $dH = 0.01$ mag (0.02 in the documentation demo script). A likelihood call is then one or two matrix–vector products per detection (times J distance nodes).

Photometric overrides. If a detection carries its own σ_m in a `mag_err` column, that Gaussian width overrides the `.eff` model for the numerator kernel of that object. The denominator still uses the block’s model for $\tilde{S}(m_t)$, because that is what the survey used when deciding characterisation.

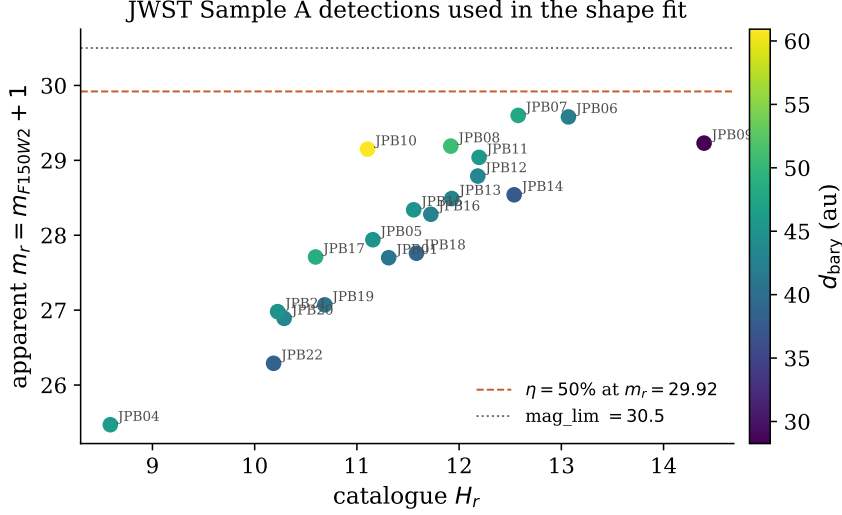


Figure 6: JWST Sample A catalogue H_r vs. apparent m_r . Colour is barycentric distance. The dashed line is the 50% efficiency magnitude in the `.eff` file; the dotted line is `mag_lim`.

Zero-error switch. `--measurement none` rebuilds the no-scatter matrices. Use it on catalogues that are already true magnitudes (some end-to-end tests), and as a difference check against the default `model` mode.

Inference. We maximise $\sum_k \ln f_k$ (MLE), then run an MCMC with flat priors on (α, β) inside generous bounds and a derived H_o from the HT count when a normalisation plot is needed. Nonparametric bootstrap of detections *within each block*, with optional log-normal draws of the cell biases, gives a sampling-plus-bias uncertainty that does not pretend $\sum 1/P$ is a known constant.

Calibration checks. Probability-integral transforms (PITs) of the observed magnitudes under the fitted model should be uniform if the shape and the measurement model are right. Posterior-predictive p -values compare the observed PIT Kolmogorov–Smirnov distance to the distribution expected for uniform deviates of the same sample size (replicates drawn from the model at a posterior θ are uniform by construction, so we do not need to resimulate magnitudes for that particular PPC).

7 Results: JWST Sample A

7.1 Sample

Eduardo et al. (2026) Sample A provides 20 multi-epoch JWST detections. We take positions, m_{F150W2} , and d_{bary} from `jwst-tno-followup/data/jwst_sampleA.csv`, map to r with +1 mag, and build catalogue H_r with the same Bowell inversion the debias code uses. Selection and photometric parameters come from `characterization/JWST_sampleA.eff` (and the three epoch copies). Figure 6 shows the detections in the (H, m) plane.

7.2 Shape-only MLE under three treatments

Table 1 and Figure 7 report a documentation-demo fit (the standalone script in this directory; the production CLI will write the same comparison into `hfit_summary` once the measurement-error branch is merged). Three treatments are shown:

Table 1: Shape-only MLE on JWST Sample A ($n = 20$). Bounds were $\alpha \in [0.05, 2]$, $\beta \in [0.02, 2]$. In all three treatments α pegs at the lower edge: with this sample size and magnitude distribution the conditional likelihood does not constrain the bright-end slope.

treatment	α	β	H_B	$\ln \mathcal{L}$
no measurement error	0.05	0.251	11.25	-29.695
photometric model	0.05	0.249	11.28	-29.694
photometry + 5% distance	0.05	0.256	11.24	-29.681

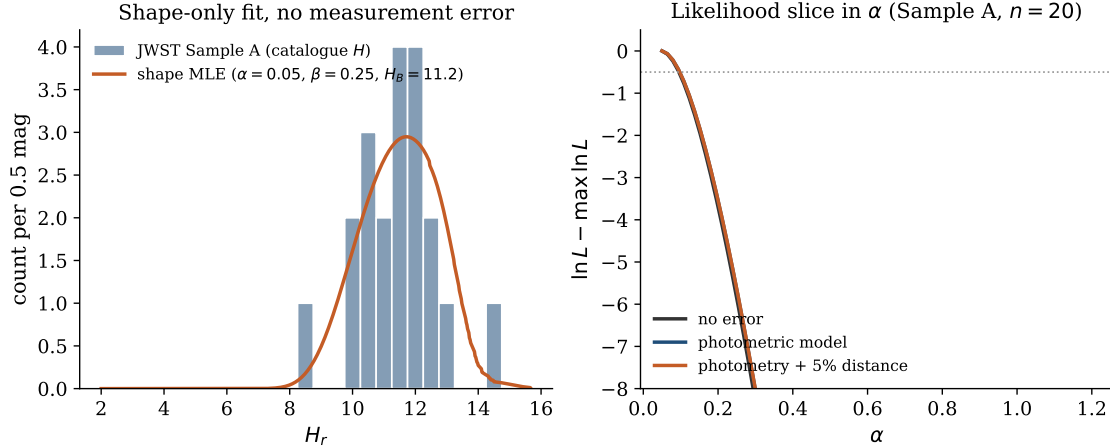


Figure 7: Left: catalogue H histogram and the no-error shape MLE prediction. Right: likelihood slice in α at the no-error MLE's (β, H_B). The three measurement treatments barely separate; all prefer the smallest allowed α . That is a sample-size / leverage statement, not a claim that measurement error is irrelevant in general (cf. Figure 4).

1. no measurement error (equation (6));
2. photometric model from the `.eff` file, fixed distances;
3. photometric model plus a illustrative 5% fractional distance error on every object (seven distance nodes).

7.3 Reading the result

With twenty objects, most of them piled up around the efficiency knee, the conditional shape likelihood is almost flat in α once a taper is allowed to eat the faint end. The MLE therefore sits on the prior edge $\alpha = 0.05$. Measurement error—at the level encoded in the current `.eff` file, and even with an imposed 5% distance error—does not move that conclusion. That is expected: Sample A's `mag_error` line was shifted to ~ 30 mag and is small on the bright side of the drop, and distance errors of 5% are sub-dominant to the Poisson noise of $n = 20$.

The honest summary for Sample A today is therefore:

- The *machinery* (no-error vs. photometric vs. distance mixture) is in place and has been checked on synthetics (Figure 4).
- The *data* do not yet determine α ; any published slope from this sample alone would be prior-dominated in a shape-only conditional fit.

- A useful next measurement is the HT-normalised cumulative counts with bootstrap bias errors, which ask a different question (yield, not shape), and/or a joint fit with a larger ground-based sample sharing the same $\lambda(H)$.

Numbers in Table 1 were written by `make_figures.py` to `results/jwst_shape_mle.csv`. Re-run that script after changing the `.eff` file or the detection table.

8 What we are still not doing

- Colour and per-block zero-point nuisance parameters with priors. Those are shared offsets, degenerate with H_o and H_B .
- A population distance prior $p(d \mid i)$ from the OSSOS model, needed before distance marginalisation is really Bayesian for `model_ae` surveys.
- An extended Poisson likelihood for the normalisation, replacing the HT count. Until then, HT cells are still labelled by measured H while the bias plant uses true H .

9 Reproducing the figures

From the repository root, with `numpy`, `scipy`, and `matplotlib` installed, and with `../jwst-tno-followup` (or `/tmp/jwst-tno-followup`) checked out:

```
python docs/lf_shape_fit/make_figures.py
pdflatex -output-directory=docs/lf_shape_fit \
        docs/lf_shape_fit/lf_shape_fit.tex
```

Figure PDFs land in `docs/lf_shape_fit/figures/`; the JWST MLE table lands in `docs/lf_shape_fit/results/`. When the production `hfit` CLI grows the `--measurement` flag, this note should be updated to point at `hfit_summary.json` instead of the demo script's CSV.